Thomas (Chenghao) Wu

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EDUCATION

Cornell University *Aug 2024 – Dec 2026*
M.S. Systems Engineering *Ithaca, NY*

University of Shanghai for Science and Technology *Sep 2019 – May 2023*
B.S. Double Major in Business & Finance and Information Science *Shanghai, China*

SUMMARY

AI engineer building LLM, RAG, and multi-agent systems with evals and guardrails, deployed to real users. Built a 6-agent purchase-requisition MVP with Claude Code at a pharmaceutical company and a RAG-based AI grading assistant serving 200 Cornell students. Strong in Python, SQL, REST APIs (FastAPI), Docker, and cloud infrastructure (AWS/Azure, CI/CD).

SKILLS

GenAI & Agents: LLM APIs (Anthropic, OpenAI; tool/function calling), RAG, vector search (pgvector), retrieval evaluation, multi-agent orchestration, MCP, LLM evals, guardrails, Claude Code, Cursor
Languages & Frontend: Python (pandas, NumPy), SQL, JavaScript/TypeScript, React, Bash/PowerShell
Data & Pipelines: Airflow, dbt, Snowflake, PostgreSQL, MongoDB, Redis, ELT design, dimensional modeling
Serving & Cloud: FastAPI, Flask, Django, REST APIs, Docker, GitHub Actions, Terraform, AWS (S3, Redshift, SageMaker), Azure, Posit Connect

EXPERIENCE

Otsuka Pharmaceutical Companies (U.S.) *Jun 2026 – Aug 2026*
Software Engineer Intern *Princeton, NJ*

- Built a 6-agent purchase-requisition MVP in 8 weeks with the procurement and Center of Excellence teams: wrote agent specs and tool definitions, generated the implementation with Claude Code, showed the first working demo in week 3, and reached ~90% accuracy on labeled and synthetic test cases.
- Ran a spec → build → demo loop across 3 stakeholder demos; for edge cases users surfaced (e.g., pending requisitions from departed employees), updated specs and added guardrails: flagged, failed-validation, and low-confidence requests go to human reviewers, and their corrections feed back into the test set.
- Built a Snowflake ELT pipeline (dbt with schema tests, stored procedures, JavaScript UDFs, scheduled tasks) replacing a manual SAP Ariba reporting process, saving ~10 hrs/week.
- Automated report distribution with Python and PowerShell-driven Power Automate flows calling Snowflake APIs, saving ~5 hrs/week.

Lily Apparel Holdings Inc. *May 2025 – Dec 2025*
Software Development Engineer Intern *New York, NY*

- Owned delivery of an ingestion platform replacing 12 manual processes: Azure infrastructure via Terraform, Docker pipelines, and push-triggered ingestion, testing, and deployment via GitHub Actions.
- Designed a modular Python/Selenium backend, shipped 8+ REST endpoints with payload contracts agreed with dashboard consumers, and cleaned and normalized large volumes of cross-platform data with pandas before ingestion.
- Built an HTML dashboard for design and marketing teams tracking user retention and view counts across external platforms, backed by Python REST APIs and deployed on Azure.
- Fixed duplicate and dropped records under network instability with idempotent processing (unique transaction IDs, retries), reaching 99.9% reliability.

PROJECTS

AI Grading System — Cornell SYSEN 5300 *Jan 2026 – Present*

- Built a RAG pipeline on pgvector that chunks course materials by chapter and retrieves grading rubrics and course content via top-k vector search, reaching a 95% retrieval hit rate.
- Achieved 99% agreement with human graders in shadow testing on 2,000 submissions (pre-set threshold: >95%) for an AI grading assistant built with Cursor, served via FastAPI REST endpoints and deployed on Posit Connect to 200 students; edge cases route to human review.
- Built teaching-mcp, an MCP server brokering Canvas permissions, exposing tool-calling access to the grading assistant.
- Built an LLM gateway with concurrency and token-rate throttling, per-call cost estimation, and fail-closed hard spend caps; orchestrated grading with Airflow DAGs (scheduled polling, LLM evaluation, retries, failure alerts).
- Built a resumable, idempotent eval harness across 17+ model configurations that recovers from mid-run failures without duplicate cost.

EBus Fleet Charging Optimization Platform — Cornell CTECH *Aug 2024 – Dec 2025*

- Built an enterprise React dashboard for fleet operations on Django/Flask REST APIs, with vehicle operations data stored in MongoDB and PostgreSQL; added Redis caching for vehicle-status queries that were re-fetched on every refresh.
- Primary technical liaison with TCAT: ran requirements sessions with dispatchers, shipped their workflows as features, coordinated 10+ contributors.
- Presented the project to TCAT's board as lead presenter, leading to an approved pilot and follow-on collaboration.